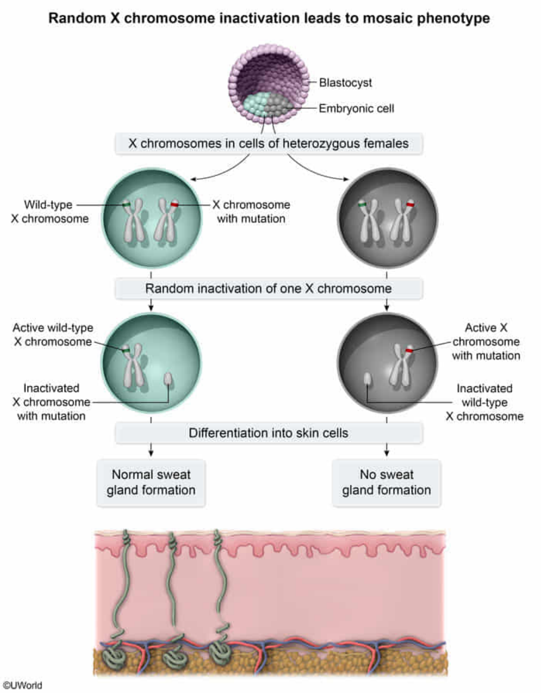
A specific X-linked mutation prevents the formation of sweat glands within the skin. Individuals heterozygous for this mutation exhibit a mosaic phenotype, with some patches of skin lacking sweat glands and other patches containing a normal distribution of sweat glands. Which of the following statements best explains the genetic mechanism underlying this mosaic phenotype in heterozygous individuals?

- A. Expression of Y chromosome genes masks expression of the mutant gene. [14%]
- B. Genes on paternally inherited X chromosomes are silenced. [14%]
- C. Mutant genes on the X chromosome are never expressed. [3%]
- ✔ D. One X chromosome is inactivated at random in embryonic cells during development. [67%]

Correct
67% answered correctly

Explanation:



In mammals, sex is determined by an inherited combination of sex chromosomes. Females (XX) inherit one X chromosome from their mother and one X chromosome from their father. Males (XY) inherit an X chromosome from their mother and a Y chromosome from their father. **Sex-linked** genes, located on either the X or Y chromosome, exhibit different inheritance and expression patterns between males and females. Specifically, females inherit two copies of X-linked genes whereas males inherit only one.

The question describes a recessive X-linked mutation that prevents sweat gland formation within the skin. Heterozygous individuals (ie, having one mutant X-linked allele and one wild-type X-linked allele) exhibit a mosaic phenotype in which some skin patches lack sweat glands and other patches have a normal distribution of sweat glands. Because this gene is X-linked, **only females (XX)** carry two copies of this gene and can be **heterozygous** for the X-linked allele.

The mosaic phenotype in these heterozygous females indicates that **some skin patches express the mutant gene** and **other skin patches express the wild-type gene**. This is best explained by **random inactivation** of **one X chromosome in embryonic cells during development**, which generally occurs in mammalian females.

Depending on which X chromosome is inactivated, some embryonic cells will contain an active copy of the wild-type X chromosome, and some will contain an active copy of the X chromosome with the mutation. Accordingly, some of these cells will divide and differentiate into patches of skin cells exhibiting either wild-type expression (normal sweat gland formation) or expression of the mutation (no sweat gland formation).

(Choice A) Because the individuals described in the question are *heterozygous* for the *X-linked gene*, they must be *female* (have two X chromosomes) and would *not* carry a Y chromosome.

(Choice B) Silencing of paternally inherited X chromosome genes would cause the expression of *only maternal* X chromosome genes in *all* cells of the female organism. For females heterozygous for the mutation, this would cause expression of a single (*not* mosaic) phenotype.

(Choice C) If mutant genes on X chromosomes were never expressed, the individuals with this mutation would display *only* the wild-type phenotype and exhibit normal sweat gland formation.

Educational objective:
The inheritance and expression of sex-linked genes (located on either the X or Y chromosome) differ between males and females.

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